

# Thomas Parr: The neurobiology of active inference

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Technical background **Duration:** 49:53 | **Uploaded:** Unknown **Transcript**

**method:** auto\_caption

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um it's a great pleasure to introduce uh thomas parr today um very briefly about his background so he started medical school in 2012 i guess yeah at the ucl and did did also like a bachelor there with a focus in new science in he started his p i hope it's uh everything is correct he started his films working working together with carl fristen and darren reese um so i think thomas is really like the person who most frustrated me i mean i was at the at the phil yeah so a long time ago i was fascinated by like the idea of the free energy principle and uh carl tristan and really started going uh learning about boltzmann equations and uh some kind of like message passing and so on so forth yeah and over years and years and then in 2016-17 i realized that uh suddenly a number of papers were coming out which became much more accessible and on these paper there were always like one name yeah too so i looked into that and i realized that there is one person out there who is uh so young and who just started his phd when he was able to go all through this um extremely complicated um math and ideas uh so easily yeah that i thought i should really give up thinking about the free energy principle at all and more impressively um i think a couple of months ago even like a book came out like um about like active influence uh where he is like the first author on um and i really can recommend this book it's a it's a fantastic introduction to active inference to the free energy principle so it's a really really great pleasure to have you here tonight uh thomas and i'm really looking forward uh to your talk yeah um thanks for joining yeah well thank you very much for the very nice and the well the invitation and the very nice introduction as well um first of all can you hear me okay yes okay and next thing is to see whether i can successfully share my screen uh let's try this so i'm assuming you can see the powerpoint and now hopefully in full screen perfect great so today um i realized that you've already had a talk from ryan smith previously so i'm assuming he's given an excellent overview of active inferences as he always does um and i wanted to take the opportunity to focus in on something a little bit more specific um in in this talk which is thinking about how how dealing with active inference lets us think about neurobiology and how we can start to make connections and form neurobiological

theories from um from the principles and so i'll start just by giving an introduction that i i think will hopefully give everybody who a little bit of a refresher on active inference and the basic principles um and although there are some technical elements to it i'll try to make it as intuitive as possible and it really doesn't matter if you get all the technical detail through this it's just to hope to build some intuitions and so i i tend to like to start talks with this slide which just shows two different sorts of system the one on the left is one that just gradually diffuses over time sort of loses form becomes progressively less interesting whereas the one on the right despite having the same amount of randomness built into it manages to maintain its form over time and a useful starting point in thinking about active inference is thinking about what the difference between these two sorts of systems are and how how the one on the right is able to maintain its form and resist this diffusive um the effect of these random fluctuations that are forcing it in all sorts of different directions um and it's often useful to reformulate this in terms of in terms of the probability density or the change in the probability distribution of all of these little particles over time so again you can see on the left we've got something that's just diffusing out into nothing and on the right we get something that behaves from a proper ballistic level pretty much statically um and it's really those on the right that we're interested in those biological creatures are able to maintain their form are able to resist the effects of what the environment does to them now we if we sort of write down the kinds of dynamics that we need for a single particle or single part of the system to try and maintain this form what we get to is is a system that i think quite intuitively always moves when it can from regions of low probability to regions of high probability so we interpret this final distribution it gets to as some steady state all we need to do to maintain that in a random environment is to always try and climb up hill of the probability gradients and then almost by definition we end up spending more time in highly probable states and less time in more improbable states and this is sort of the starting point really for active influence because we now have this notion that we're climbing probability gradients and we have some distribution that we're effectively trying to maximize and we can link that back to ideas like the bayesian brain the idea that the brain is using some model of the world around it to generate predictions um and and then is drawing inferences about the data that it actually obtains so sensory information coming in through our eyes ears through our skin that we can then use to draw inferences about the causes of those data um so here we've got the idea that there's some states of the world  $x$  that are causing some sensory data why we're then forming some posterior beliefs so beliefs about the causes given the data and we're doing this using a generative model that comprises a

likelihood and a prior so prior being how plausible are the things out there in the world before we've made any observations the likelihood being how likely the observations we've made are given the states of the world or given our hypotheses about what caused them and together we refer to these two things as a generative model now the posterior distribution is what happens when we invert that model where we find the probability of some causes given the data and the final term we've got at the end here is referred to either as an evidence or a marginal likelihood and that's because in the context of bayesian statistics we often use a marginal likelihood as a measure of the fit of the model to the data that it's trying to explain because how much evidence do those data afford some model of the world and together we can think of these as being an inversion of generative model now the reason i put this here is that we can see that the dynamics that i'm showing in the upper left can be interpreted as the process of maximizing the evidence of some model maximizing the fit between the brain's model of how the world works and how the world and then engages with the model or engages with the brain by presenting it and broadly there are two ways of doing that the first is as we've already spoken about is changing your beliefs based upon new data such that you get a better fit to the world the other way which i think is one of the key ideas in active inference is that you generate actions based upon your beliefs that change the world to make it more like your model and this is sort of the key idea that underwrites active inference that that all we're trying to do is maximize the evidence as a model of our world and we can either do that through perception by changing our model or through action by changing the world but together they come under one um one single objective sometimes referred to particularly by people like the acapella as self-evidencing now i want to go a little bit into the structure of generative models and specifically the ways we can think about and the ways we can notate generative models because it often helps to move from a slightly more mathematical abstract description to a more graphical notation that i think often gives a much better intuitive sense of what's going on so to do that i'm going to start with this idea that model evidence is effectively what we get out once we've integrated out all of the causes and from our model uh by which i mean if you take account of all of the things our model predicts and then we take into account the prior probabilities of all of the all of the things that are causing um those data uh we can then work out what the probability of the data are under that model but the model the model itself takes account of the things that are being generated so our sensory data and also the things that are causing them simply a joint probability distribution involving all of these things so i'm now interpreting this graphic up on the left as and depending upon some

generative model where we've effectively integrated out everything that we don't need to determine  $u$  explicitly now generative models will often have some interesting structure and in fact if they don't have interesting structure then they're not interesting generative models and normally that structure manifests as a factorization of this joint distribution but not everything depends upon everything else  $u$  and so we can often factorize it and here I'm just showing a completely arbitrary factorization of some general model  $u$  where we have certain dependencies so  $y$  depends directly upon  $x_1$  and  $x_2$  but not directly upon  $x_3$  and so on and the reason it's useful to think about this factorization is because we can then express a graphical version of this model that provides a bit more intuition as to what's going on so the way we do that or one way that we can do that is using something known as a factor graph and the idea is that we take each of these factors in turn middle or square and then we draw an arrow to whatever whatever's on the left of the probability  $u$  factor we're interested in so here an arrow towards the  $y$  because we're saying the probability of  $y$  conditioned upon or depending upon the other two things and then we attach those to the same square we then move on to the next factor for another square and we then attach the relevant variables together in exactly the same way and we carry on doing that until we have a picture of what our model is like so now instead of having to look at this equation you can look at this and say okay well  $x_3$  causes  $x_1$   $x_4$  causes  $x_2$  and together  $x_1$  and  $x_2$  relate to or together generate our data  $y$  this is just to demonstrate several models you may be familiar with generally and the fact that we're often implicitly using these kinds of generative models without necessarily thinking about it so when we're performing a principal components analysis we're often taking some some distribution of some variable  $x$  we're then mapping that to a higher dimensional space  $y$   $u$  and then and that's our sort of model of how the data that we're working with are generated so what we can then do when we perform a principal components analysis is take our high dimensional data  $y$  work out how you go back from the  $y$  to the  $x$  and that gives you all of your specific principal components and the directions that have been stretched in various different ways same principle applies for things like canonical various analysis where you've got some set of  $u$  variables then mapped to two different sorts of data by stretching them and and distorting them in various ways and the challenge is how do you get back to the the common  $u$  hidden variable common cause for both of those data sets and we do the same sort of thing with something like clustering analysis where here the model says we have we're effectively sampling from one of several different clusters with different probabilities  $u$  we're then generating some point in space depending upon which cluster it falls in and the process of

um of inference is again undoing this it's finding the posterior by taking a point in the cluster and saying well which cluster is that from so going from the wide to the s so that's a sort of brief introduction to the idea of generative models the um the role they play in active inference some key examples and some of the notations i'm going to use as we go forward what i've said so far has been relatively abstract and in the next section i want to try and work from those sort of general principles and these slightly abstract notions through to something more specific and more neurobiological the first thing we're going to do to get there is to introduce one more abstract concept which is that of a markov blanket a mark of blanket comes up in all sorts of places in active imprints but i'm going to use it in a very specific way here and and this is to talk about conditional independence and what i mean by conditional independence is that if you have if you have two different sets of variables so here i've notated them as the  $\mu$  and the  $\eta$  the markov blanket  $b$  is the set of set of variables that render the other two completely independent of one another that mean that if you know everything about the blanket knowing something about  $\mu$  tells you nothing new about an  $\eta$  to give you an example of this one of the examples that most people are familiar with is the idea of a markov chain where you have a sequence of events in time so something that happens in the past then influences the present which then influences the future with no direct influence from the past to the future without going via the present so if i know everything there is to know about the present knowing something about the past tells me nothing new about the future and vice versa so in that sense the present is the markov blanket that separates the past from the future now this is a very useful concept when we're dealing with graphical models or generative models that have some interesting factorization structure and i'll try and explain why the markov blanket is often referred to as uh or can often be identified by identifying variable we're interested in so let's say we're interested in the variable  $x_2$  then you find the parents of  $x_2$  so the things that caused it the children with  $x_2$  so the things it causes and the parents of its children and that that gives you the markov blanket of  $x_2$  now the significance of this is that knowing this means that if we know about the blanket it doesn't matter what else is going on in the generative model you know this might be some tiny section of a huge generative model with many many variables in but we don't need to know about those variables all we need to know about are the blanket variables so we can effectively ignore everything else if what we're interested in is  $x_2$  now how does this then matter the neurobiology well i think the answer to that is that the brain is an extremely sparse structure that synaptic message passing does not involve connections between every different neuron in the brain to every other neural brain

there are a small number of relatively speaking of um synapses between any neuron and um and its neighbors or or neurons elsewhere in the brain and so we can make an argument that if the brain is performing inference without lots of different variables all it needs is to know the the structure of the generator model and the variables that sit in that markov blanket or at least the neurons that are representing those other variables and that tells us that we can then when we're inverting that mole when we're performing inference what i'm showing here is just one example of an inference scheme we can we can express the uh the dynamics of that neuronal message passing in terms of connections from the other neural populations that are representing the variables in that markov blanket the details of this aren't that important but it this is one example of a an inference scheme known as variational message passing here formulated is a gradient scheme in terms of some dynamics um which has relevance for um for modelling neural populations where interest and how the dynamics of those populations evolve over time um and we can sort of do this recursively we can couple together lots of different parts of this system where the things that have been connected up are are just the mark of blankets of them of different populations here i've expressed it in terms of an error term that this epsilon um which effectively represents the gradient of the free energy objective which is used as a an approximation to that marginal likelihood or evidence that i spoke about earlier and then some beliefs about or expectations about um that variable so we can now use our the errors in our predictions to update those variables and once we've written down the dynamics of this um sort of system we can simulate things like firing rates we can simulate the dynamics and the behavior these networks of neurons passing messages between one another um as a means of performing a form of inference thus uh self-evidencing i know this is probably still seeming relatively abstract but i think what will hopefully help is if we then go through some examples thinking about specific kinds of generative model and how that might then affect um the kind of message passing we would see in a brain and the first step to thinking about that which i think is very important when we think about really useful generative models is they will all typically have a temporal aspect to them but as biological creatures we deal with things that evolve in time and so as we're thinking about how do you formulate a generative model that has that kind of dynamic aspect to it and there are several different answers to that and and one of them is we we use something known as as a taylor series approximation um so we start by saying okay at the current time what what is the value of some variable  $x$  in the world and this may be some continuous variable it may be where my arm is in space if we then want to know how it's going to evolve in time we could then say

well let's take the next element of our taylor series approximation let's take account of its velocity and that tells us a little bit more about the trajectory that my arm might be on we can then take another value which is the current acceleration of that um of that position and we get a slightly better approximation to the trajectory and the more terms we add in the greater approximation we have of the trajectory of the greater representation we have in the trajectory just as a series of numbers uh where those numbers are my current position my current velocity acceleration etcetera so we could formulate a generative model by trying to predict each of these coefficients of this taylor series or each of these generalized coordinates of motion as they're sometimes referred to an alternative is we just say at time one where will i be at time two where will i be and we simply represent it in terms of the sequence of points over time often uh when we're discretizing time in this way we often we also discretize space so we might say okay where am i in some discretized scheme am i in location one two three four or five at each different time points um and so we have two two different ways of accounting for how things might evolve over time and each of these can be useful in slightly different circumstances so if i were dealing with a sequential decision-making task it may be much more efficient for me to say okay the first move i'm going to make is this then the second one is this the third one is this and dealing with something very sequential that might also be important in in sort of language processing where you have to think which words you're going to put together in a sequence and we're dealing with discrete variables over time because words are a categorical entity that's supposed to continue on as opposed to lying on a continuum however if i were dealing with movement or the responses i'm getting from some photoreceptor in my retina it may be much more sensible for me to be working in continuous time thinking about how things evolve at a continuous scale so what i'm showing here is an example of a generative model and the associated message passing scheme when formulated in terms of this continuous time scheme so what we've got here is a factor graph at the top this blue element that uses exactly the same sort of formulation that we've been discussing so far so we have some variable here which might represent our our um position which then predicts our velocity at that time which then predicts the acceleration at that time so we have a series of predictions here about the coupling between different orders of motion at each point those are predicting some sort of data here are y's which represent the current position of our sensory data the current velocity the current acceleration what i'm showing lower down here is the message passing scheme we get when we take account of the markov blankets of each of these variables so the markov blanket here for the the velocity would include the position and the

acceleration but also the data i've got about those things um and some prior beliefs about about other variables in our model and so we'd end up connecting those things up and although it looks like a bit of a mess of connections it's still fewer connections than the total number connections you would get if you map everything to everything else what we've got here takes the form of effectively a predictive coding style scheme um which people may be familiar with and the idea that you can you can hierarchically predict um predict some data and you can predict the thing that's predicting that data those data and by observing the errors in your prediction of the data you update your prediction which then cascades up a hierarchy allowing you to update subsequent predictions what i'm showing you here is the equivalent model formulated in terms of discrete time and in this case discrete space as well so here we have a state at time  $t$  minus one which generates a state at time  $t$  which generates a state at time  $t$  plus one at each of those time points we're generating some observable outcomes which are represented as the  $o$ 's i have a variable  $\pi_i$  here that represents alternative trajectories  $i$  could  $i$  could pursue so alternative policies or plans alternative ways  $i$  could change the sequence of events and again we have um underneath the message passing scheme that could perform inferences about this sort of um and you can see that the message passing scheme to some extent looks like an inversion or or as loosely a sort of mirror image of the generative model because all it's doing is taking each step that was performed to generate some data and then inverting those steps one at a time so it's just going backwards from the data to arrive at the causes of those data now what i've got shown here on the left is just a sort of cartoon image some of the key connections in cortical micro circuits so here we've got superficial granular cells as  $u$ h as  $sp$  uh spiny stellate cells as  $ss$  deep pyramidal cells as  $dp$  and inhibitory interneurons is the double eyes um and this is clearly an over simplification of what is a very complex connectivity structure but the reason i'm showing this here is that if we know something about this connectivity structure we know about some of the key patterns we can start to ask how can the the sorts of models that we've spoken about over the last couple of slides then be interpreted in terms of the micro circuits that could help um the could help implement these in a biological system so what i'm going to show in the middle is an example of a continuous states-based predictive coding style model has all the same elements as in the slide i showed you before that are now mapped so that some of the connections coming in and out of it um loosely map to those associated with a known cortical anatomy and we can do exactly the same thing here with the discrete state-space model and think about how we can assign those roles um associated with roles of different cortical layers and and then use that as



a way of forming hypotheses now what i'm showing you on this slide is not it's not necessarily the last word on this and you could formulate several different hypotheses about how these two things map together and that's where a lot of interesting science happens this is just one example of how associate what we know about cortical microcircuitry and anatomy with what we know about the anatomy of message passing for some simple generic forms of generative model what i'm going to try and do over the rest of this presentation is to take the connections coming out of this out of each of these um points and to try and think about how those might um all the roles there might those might play when we take an active inference approach and the sorts of structures in the brain that might be involved in uh dealing with those dealing with those things i'm going to start by talking about movement and reflexes so here the kinds of things we're interested in are the connections that leave the cortex and go to areas like the spinal cord and the motor neurons the predictions that we can we can then make about the um what we're seeing here is the in the central plot and the g that's been circled in red is a prediction about the sort of data i might expect to observe which may here be proprioceptive data consequent on particular sorts of action and we can think of this quite simply if we think about the structure of a reflex arc so um what i've got here on the lower part is just the the same marginal likelihood or model evidence that we've dealt with before this idea of it being a measure of the fit of a model both in terms of its accuracy the accuracy with which it predicts some and and the simplest least complex version of the model that will work for that what do we do when we change the data well we try and make the data more accurate in relation to our model by by trying to align the data with our so when we think of the structure of a reflex arc where we can interpret it as some descending prediction coming from the motor cortex saying what is the proprioceptive data i expect to observe by comparing that with the proprioceptive data that's actually coming into the spinal cord through the dorsal horn with that prediction we then get a potentially a mismatch between the two that can be used to generate action by the ventral horn generating movements and contractions and muscles that then um try to fulfill these i'll show you just an example of of a simulation of an arm using active inference that has these sort of reflex arcs built in and all we're doing here is we're just tapping on the biceps tendon and where the red arrow is pointing and we're just seeing the reflexive response that we're getting as a consequence of effectively inducing this additional piece of sensory proprioceptive information by stretching that tendon we induce the mismatch between the prediction that's coming down and the um and the data that's coming out and that man that is corrected then by generating this additional movement interestingly we can do things by

manipulating the confidence of those predictions and sometimes get things like bigger reflexes and hyper reflexes the sort you might see in clinical populations with spinal cord injuries and and so you can see the reflex is slightly larger than it was before and slightly brisker uh i'll show you another example of this same idea where now we've developed a mapping between the micro circuit and the anatomy of the oculomotor brainstem so here starting from the superior colliculus the the rough the rostral uh interstitial nucleus of needle longitudinal vesiculus and various other centers and and the cranial nerves that are responsible for generating eye movements so cranial nerve three and cranial nerves and six here um are seen here as resulting from uh these error terms so we're predicting how i would expect my eyes to move um and any error in the current position of the eyes is corrected by generating eye movements so by inducing different priors on top of um on top of this model different predictions about where i'd expect the eyes to be we can actually then generate different sorts of eye movements using a predictive coding style active inference scheme where the key thing is that the predictions can be fulfilled by actually changing the positions of our muscles and positions of our eyes so that sort of provides a very brief overview and a couple of examples dealing with the role of reflexes and predictions in generating movements but it doesn't really tell us anything about how movements are chosen how movements are selected and you don't necessarily get any intelligent behavior out of that and for that we need to start looking at different sorts of structures and different parts of this generated model and here the key thing i want to focus on is the role of the basal ganglia and which here we're going to associate with the computation of something known as the expected free energy and i'll try and describe what that is a little bit more in some of the subsequent slides what i'm showing here is a mapping of some of the um subcortical but some cortical anatomy or more accurate i should say that this is a mapping between parts of our bayesian message passing scheme parts of model inversion when we're dealing with ability to plan and make decisions to some of the known subcortical anatomy of the brain and the key thing i want to focus on is this g that his hair depicted him as part of the um as part of the striatum so the things that are feeding into this uh are what we're getting from the cortex here which is some prediction of the outcomes we'd expect if i were to pursue a particular course of action which is this subscript pie and the um another sort of error term here which is how far away is is um are those predictions from my preferences about how the world should be my prior beliefs about how um the kind of data that i would actually actively seek out and act to get together those are used to calculate our expected free energy which we can then use to formulate beliefs about policies by saying that um the policies we

would select the plans we would engage in are those that we would expect to be associated with the lowest um expected free energy and to put that more formally um we're saying that we're going to give a prior belief that um the policies the series of actions we're going to choose are going to be those associated with the maximum information game tell us the most about um the world around us they'll fulfill our preferences um and then we're going to add an additional term here which deals with habitual type policies and things that we tend to do because we've learned we behave in a particular way in those situations now the combination of this information gain and preferences are often referred to as as an expected free energy and i won't go into the details here but that's simply because you can rearrange them mathematically to make them look very much like the equation for a free energy or or marginal likelihood approximation with an expectation around them to say that these are the what we would expect given um our predictions about the data we would obtain because here we're dealing with beliefs about the future plums into the future where we haven't yet had those data and we need to deal with the expectations of what those data would be under different plans that we could choose so here here we're showing just an example of the direct pathway through the basal ganglia um which is normally thought to facilitate movement and depends upon this expected free energy and an indirect pathway which here we've associated with these kinds of habitual drives and i'll come back to those a little bit later on when we deal with hierarchical models just to give you a little bit of intuition as to the information gain aspect of the expected free energy because i think most people uh most people probably understand this idea of seeking preferences and behaving to maximize some degree of reward but many people are less familiar with the idea of seeking information into the way that might manifest in these kinds of so as an example um i'm going to show you just a simple simulation where we're going to manipulate some of the different aspects of the uncertainty in the model so here um in the upper left i'm just showing a way of parameterizing um the the uncertainty associated with the data generating process so this is our likelihood precision our sensory precision and it effectively says that when this precision is very high we can be very certain about the outcome we'll observe given a particular state of the world whereas when it's very low we can predict everything with very similar uh we can do something very similar by manipulating the uncertainty in the dynamics of the world so again when this is very high it means that where i am now is very highly predictive of where i'll be at the next step in time whereas when it's very low it means that pretty much anything at the next time point is is equally probable and i'll show you a simple simulation where these two things are manipulated

just to try and give you an intuition for what it means to act to maximize one's information about the world so the upper simulation here top left shows um four panels which can each change at some point in time to a different color um with some random probabilities and the blue line here is designed to show effectively an eye tracking trace so it's a simulated agent who is allowed to choose which of these panels it wants to look at at any one time and you can see it samples them with a relatively even frequency in the middle panel what we've done is we've reduced the precision or increased the uncertainty associated with the likelihood in the lower left that effectively is like turning off the lights in that location it's effectively making that lower left location much more much less informative much more noisy and so effectively what we've got here is a system that then ignores that it says that i can't get good quality high quality information from there so i'm going to look at all of the other locations rather than rather than this in the lower left an analogy for this is if you're thinking about how you perform a scientific experiment you would probably aim to use if you had a choice between two different measuring instruments you would choose the one that gives you more precise measurements rather than the one that gives you very noisy measurements we've manipulated the uncertainty or the volatility associated with the dynamics so here what's happened is the upper left location is associated with much more uncertain dynamics so here what happens is that i end up sampling that upper left location much more frequently and intuitively this makes a lot of sense because if you've looked somewhere very recently normally you will you'll know a lot about what was there you don't need to look back there anytime soon however if it's very volatile if it changes quite uh with some degree of randomness and you have very little certainty about the state of it after you've looked away you'll look back with a much greater frequency so we've effectively decreased the inhibition of return in this location by increasing its volatility or decreasing the precision associated with its um with the dynamics in that location the aim of this was just to show you this sort of emerging behavior just by having an information seeking objective by um by having a prior belief that we're going to act to minimize some expected free energy one component of which is to maximize our information about the world now the next thing i want to come on to is the role of hierarchical generative and one of the key benefits of having a hierarchical model is that we can now deal with things that evolve over a range of different time scales we might have some things that evolve very slowly and some things that evolve very quickly and to some extent we can separate out the two and use slowly evolving things to to to help us predict what's happening at the faster level and here the key things to denote are all of these connections between higher cortical regions

and lower cortical regions which manifests both in the discrete and continuous state space models and so it's worth them thinking about what the role of these are and how that manifests in terms of the generative models we've been dealing with before and it's really this that links back into the idea of predictive coding as many people know it and you've probably seen graphics of this sort in the past where we have a range of cortical regions that are each making predictions about the others and so here going from going from the right to the left we've got a series of predictions and as these dark black lines with prediction errors passed back up using these dashed lines that then allow us to correct at each level and all this this graphic shows is a simplification of the graphic on the previous slide so i want to give you an intuition for why this is useful using a couple of examples so the first example i'm going to use is one from the domain of active vision um so imagine imagine you're doing a task now where you have to fixate on this cross in the center and i'm and maintain that fixation and show you a stimulus but maintain fixation on the cross stimulus is going to disappear and then your next task is to perform an eye movement to the location where the stimulus appeared it's a very simple task of the sort was used frequently in monkey electrophysiology and all throughout neuroscience but it's an interesting one because it has several different components to it that i think help in terms of thinking about the utility of one aspect of this is making decisions about where you're looking at each of those time points and making inferences about which of several alternative locations you're going to perform an eye movement too so simple form of planning here so that calls into into action the sort of discrete states-based model that we were dealing with before this partially observable decision process this set of discrete sequential time points and that's often very highly relevant structure of most simple tasks in cognitive neuroscience however we also need to actually move our eyes we need to uh we need to uh move our eyes from one location to the next we need to know how to generate forces we need to know how to make sure the eyes come to rest in the right location given those decisions and that calls them to play this sort of continuous state-based predictive coding style model so how do we then how do we then combine the two how do we deal with the situation where we want to predict a particular location at a discrete level one of several alternatives but then also map that to a continuous motor trajectory and this is where generative models get a little bit more complicated so taking this a bit at a time what we've got at the top here is our markov decision process model this is our discrete states-based model in discrete time now at each of those time points we're no longer predicting data specifically we're now predicting a short trajectory formulated in terms of our continuous state-space

models so at time one we're now going to predict a short element of our trajectory  $u_m$  and this is very much like the sort of clustering model I showed you much earlier that you take a discrete point and you map it to a point in continuous space with some at the next time we then predict the next bit of that continuous trajectory so our priors for our continuous model are now inheriting from the predictions from the discrete state space model and then we can do the same thing again at the next time point and so we can create a discrete sequence from our  $u_m$  our sequential model and at each point in that sequence associate that with a short trajectory in continuous space that then predicts our continuous data allowing us to then predict the proprioceptive information we would expect to get from the eyes when we're performing a task of this sort and you can see that that we now develop a hierarchical structure also in thinking about the relationship between the bits of the message passing scheme so here we've got the discrete bit of our message passing scheme our discrete micro circuit and a bit of the continuous micro circuit here and the interactions between the two of those we're making predictions from the discrete one to the continuous one and then passing arrows backwards to allow us to update the  $u_m$  discrete element interpreting this again neurobiologically imagine that say we have some element of the cortex perhaps the frontal line fields that's making predictions about one of several alternative locations I could I could direct my gaze we can then map that via sort of output of the basal ganglia and the predictions about what policy I'm going to pursue through to areas like the superior colliculus which then might make use of predictive coding style networks of the sort depicted here to then generate eye movements as we've seen before and just to show you that in action we imagine that these are population of neurons that at each time point trying to predict one of several  $u_m$  in this case three different locations at four different time points we can then map that through uh structures like the superior colliculus and this is supposed to sort of cartoon the idea of a population code in the superior colliculus that then results in the eye movement to each of those discrete locations putting it all together we can then formulate exactly the tasks that I asked you to perform earlier and see the result of an active inference scheme performing that task and so here you can see it performs it very successfully by predicting sequences of eye movements that are conditioned upon the part of the model and the idea of maintaining this belief about where the stimulus appeared and using that then to direct my policy selection and eventually the movement I select I want to give you one more example of this sort of hierarchical scheme in the domain of motor control and to appeal to the same sort of simulations we were looking at earlier in terms of the reflexes exhibited by that arm the generative model we're going to use

here and is a slightly complicated one but here effectively involves transitions between different locations that an arm could be in or a hand could be in which then predicts different locations in some continuous space and different locations that a target might be in um and that will then allow us to make predictions about about given where some target is in some continuous space uh which decision am i going to make about the alternative points i could move my arm to and so here we have the simulation where we have the target appearing as the black ball out of the out of the three balls which is going to change at various points in time and our active influence scheme is now selecting these locations translating those into predictions of continuous trajectories and making the appropriate movements such that it reaches each of these these target locations and what's shown in terms of the the graphic here is also an interpretation of message passing in terms of the known anatomy of the motor system or part of it we can perform various lesions to this and see whether they behave in the same way that we might expect patient populations to so here we've induced a cerebellar lesion um which effectively um involves a misestimation of certain precision terms certain um confidence or variance parameters and you see this sort of overshoot and this kind of a silly behavior in the way the arm behaves that mimics what we might expect from cerebellar ataxia and we can look at the higher levels of the model and actually have multiple uh hierarchical levels of the discrete scheme and here what we've done is we've induced a lesion in what might represent a frontal lobe lesion where now the model is able to perform all the movements with perfect fluency but every time there's a change of context it takes quite a while to adjust to that new context to deal with a new situation you'll see that instead of moving straight there as it was before there's a lot more hesitancy and a lot more difficulty constructing that long-term narrative because we've broken that slower hierarchical level and here a final lesion that we can introduce is one that reduces my confidence and policy selection so here here this effectively makes me very uncertain about what i'm going to do next and here we get a sort of a kinetic type picture of the sword we might expect in a parkinsonian type syndrome i realized that was quite a lot and i hope i'm not too far over time but i'll just try and summarize um briefly the key ideas that we've discussed so the the first was thinking about this idea of climbing probability gradients the idea that to maintain some form over and to persist we need to effectively be climbing these probability gradients all the time we interpreted those probability gradients in terms of in terms of bayes theorem and connected that to the bayesian brain and the sort of message passing that might be involved in forming inferences about elements of some generative model and the role that might have in cortical

micro circuitry we also discussed the role of action and the idea that that um one way to make our model better fit the world is to act upon it and one way to do that is to simply reduce any discrepancy between the predictions from that model and the and the data that's coming in and the structure of that is is very similar to the idea of a reflex arc as is as is well known throughout motion neuroscience um and the final thing we we thought about was how we can then construct hierarchical models that allow us to produce simulations and theories of that that deal with much larger scale networks in the brain and take a full sort of systems level view of how particular movements and particular behaviors are generated [Music] through the inversion of particular forms of generative model with that i'd like to thank many people who have been involved either directly or indirectly in this work um i'll mention the book again which was mentioned at the beginning and a lot of this talk was loosely following from the structure and a couple of the chapters in the book so um if anybody's interested um please do do take a look i'm more than happy to answer any questions thank you for your attention